

TAO ZHONG

 taozhong.info  tzhong@princeton.edu  [linkedin.com/in/neiltaozhong/](https://www.linkedin.com/in/neiltaozhong/)  github.com/n3il666

EDUCATION

Princeton University	08/2023 - 07/2028 (<i>Expected</i>)
<i>Degree:</i> Ph.D. in MAE (Robotics Track)	<i>cGPA:</i> 4.0/4.0
<i>Committee:</i> Christine Allen-Blanchette (<i>Chair</i>), Felix Heide, Ryan P. Adams	
University of Toronto	09/2018 - 06/2023
<i>Degree:</i> B.A.Sc. in Engineering Science (with High Honour)	<i>cGPA:</i> 3.81/4.0
<i>Major:</i> Robotics Engineering	<i>Minor:</i> Artificial Intelligence
<i>Advisor:</i> Animesh Garg	

EXPERIENCE

CAB Lab , Princeton University	2023 - Present
<i>Graduate Research Student, Advisor: Prof. Christine Allen-Blanchette</i>	
Topics: physics- and geometry-aware deep learning for scientific discovery [C6, C8, U1, U6, U7], robot learning [C4, C5, C7, U3], and generative foundation models [U2, U4, U5].	
People, AI, & Robots Lab , Vector Institute & University of Toronto	2022 - 2023
<i>Undergraduate Research Student, Advisor: Prof. Animesh Garg</i>	
Topics: differentiable physics for dexterous grasp synthesis [C2]	
Noah's Ark Lab , Huawei Research Canada	2021 - 2022
<i>Machine Learning Research Intern, Advisor: Prof. Yang Wang</i>	
Topics: OOD generalization via prompt learning [C3] and test-time adaptation [C1] for foundation models	
aUToronto , The University of Toronto Self-Driving Car Team	2020 - 2022
<i>Mapping & Localization Team Lead, Team Advisors: Prof. Tim Barfoot, Prof. Steven Waslander, Prof. Angela Schoellig, Prof. Jonathan Kelly</i>	
Topics: semantic mapping and SLAM optimization for autonomous driving.	
Shenzhen Institute of Artificial Intelligence and Robotics for Society , CUHK(SZ)	2020
<i>Visiting Research Student, Advisor: Prof. Huihuan Qian</i>	
Topics: web-based testing platform, state estimation, and control for marine robotics	

PUBLICATIONS

* denotes equal contribution.

Papers in Submission

- [U7] Dongzhe Zheng, **Tao Zhong**, Christine Allen-Blanchette. Symmetry Descent Neural Operators: From Local Equivariance to Global Consistency. *Under Review*, 2026.
- [U6] Dongzhe Zheng, **Tao Zhong**, Christine Allen-Blanchette. Homomorphic Generator Learning: Compositional PDE Neural Operators for Multi-Physics Surrogates. *Under Review*, 2026.
- [U5] Yixun Hu, Zhixuan Zhao, **Tao Zhong**, Christine Allen-Blanchette. OrbitRIR: Frame-Averaged Few-Shot RIR Synthesis for 3D Sound Source Localization. *Under Review*, 2026.
- [U4] Yixun Hu, Zhixuan Zhao, **Tao Zhong**, Christine Allen-Blanchette. CERPA: Heading-Equivariant Attention for Yaw-Invariant Few-Shot Room-Acoustic Synthesis. *Under Review*, 2026.
- [U3] Keqin Wang, **Tao Zhong**, Yulong Yang, Christine Allen-Blanchette. Equivariance Begins at the Source: Characterizing Compatible Sources for Generative Robot Policies. *Under Review*, 2026.
- [U2] **Tao Zhong**, Dongzhe Zheng, Christine Allen-Blanchette. HodgeCover: Higher-Order Topological Coverage Drives Compression of Sparse Mixture-of-Experts. *Preprint*, 2026.

[U1] Zhixuan Zhao*, **Tao Zhong***, Yixun Hu, Nathalie P. de Leon, Christine Allen-Blanchette. Neural Fields for NV-Center Inverse Sensing. *Preprint*, 2026.

Refereed Conference Proceedings

- [C8] **Tao Zhong**, Yixun Hu, Dongzhe Zheng, Aditya Sood, Christine Allen-Blanchette. Neural Field Thermal Tomography: A Differentiable Physics Framework for Non-Destructive Evaluation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
- [C7] Keqin Wang*, **Tao Zhong***, David Chang, Christine Allen-Blanchette. Local-Canonicalization Equivariant Graph Neural Networks for Sample-Efficient and Generalizable Swarm Robot Control. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- [C6] Dongzhe Zheng, **Tao Zhong**, Christine Allen-Blanchette. Topology-Preserving Neural Operator Learning via Hodge Decomposition. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2026.
- [C5] **Tao Zhong**, Jonah Buchanan, Christine Allen-Blanchette. Grasp2Grasp: Vision-Based Dexterous Grasp Translation via Schrödinger Bridges. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [C4] **Tao Zhong** and Christine Allen-Blanchette. GAGrasp: Geometric Algebra Diffusion for Dexterous Grasping. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [C3] Zhixiang Chi*, Li Gu*, **Tao Zhong**, Huan Liu, Yuanhao Yu, Konstantinos N Plataniotis, Yang Wang. Adapting to Distribution Shift by Visual Domain Prompt Generation. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.
- [C2] Dylan Turpin, **Tao Zhong**, Shutong Zhang, Guanglei Zhu, Eric Heiden, Miles Macklin, Stavros Tsogkas, Sven Dickinson, Animesh Garg. Fast-Grasp'D: Dexterous Multi-finger Grasp Generation Through Differentiable Simulation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [C1] **Tao Zhong***, Zhixiang Chi*, Li Gu*, Yang Wang, Yuanhao Yu, Jin Tang. Meta-DMoE: Adapting to Domain Shift by Meta-Distillation from Mixture-of-Experts. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Refereed Non-archival Publications

- [W2] Keqin Wang, **Tao Zhong**, Christine Allen-Blanchette. Equivariance Begins at the Source: Quantifying and Characterizing Source Distributions for Generative Robot Policies. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop on Geometric Representations in Robotics*, 2026.
- [W1] **Tao Zhong** and Christine Allen-Blanchette. Geometric Algebra Grasp Diffusion for Dexterous Manipulators. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop on Equivariant Robotics*, 2024.

Patents

- [P1] Zhixiang Chi, Li Gu, **Tao Zhong**, Yuanhao Yu, Yang Wang, Jin Tang. Systems and Methods for Artificial-intelligence Model Training Using Unsupervised Domain Adaptation with Multi-source Meta-distillation. *US Patent Application No. 17/966,568*.

AWARDS & HONORS

Lambda Research Grant	2026
Princeton MAE Second Year Departmental Fellowship (2 / 25+)	2024
Princeton University First Year Fellowship in Natural Sciences and Engineering	2023
NeurIPS 2022 Scholar Award	2022
SAE Autodrive Challenge: 1st Place Winner (As a team)	2020, 2021, 2022
University of Toronto Dean's Honours List (All 8 terms)	2018 - 2023

INVITED TALKS

Vision-Based Dexterous Grasp Translation via Schrödinger Bridges
GRASP Lab, UPenn, Oct 2025

TEACHING

MAE 206 Introduction to Engineering Dynamics (Undergraduate)

Spring 2026

Teaching Assistant, Princeton University

MAE 433 Automatic Control Systems (Undergraduate)

Fall 2025, Fall 2026

Teaching Assistant, Princeton University

MENTORING

Jonah Buchanan (with Christine Allen-Blanchette)

2024 - 2025

David Chang (with Christine Allen-Blanchette and Kevin Wang)

2024 - 2025

Kaison Fong (with Christine Allen-Blanchette)

2025

Zhixuan Zhao (with Christine Allen-Blanchette)

2026

SERVICE

Conference Refereeing

AAAI Conference on Artificial Intelligence (AAAI)

2027

International Conference on Machine Learning (ICML)

2026

Neural Information Processing Systems (NeurIPS)

2025 - 2026

International Conference on Learning Representations (ICLR)

2025 - 2026

Annual Learning for Dynamics & Control Conference (L4DC)

2024, 2026

Journal Refereeing

IEEE Robotics and Automation Letters (RA-L)

2025

Transactions on Machine Learning Research (TMLR)

2025 - 2026

SKILLS

Programming Languages:

Python, C/C++, MATLAB/Simulink, Bash, SQL, Assembly

Libraries & Tools:

PyTorch, NumPy, OpenCV, scikit-learn, ROS, Git, Docker, $\LaTeX$